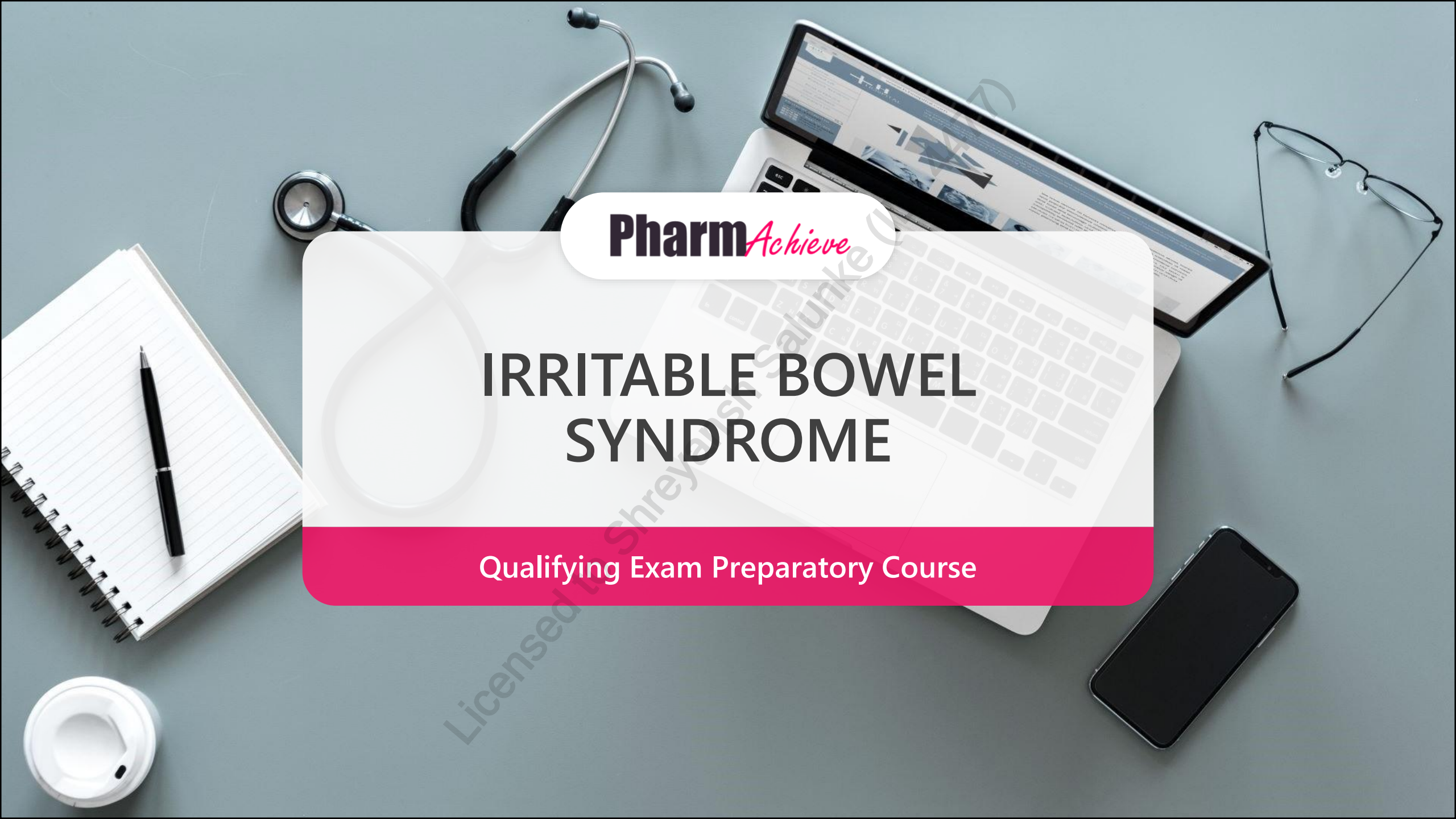




Pharm*Achieve*

IRRITABLE BOWEL SYNDROME

Qualifying Exam Preparatory Course

COMPETENCIES COVERED IN THIS PRESENTATION...

**Providing Care:
Clinical Care**

**Providing Care:
Distribution**

**Knowledge
& Expertise**

**Communication
& Collaboration**

**Leadership &
Stewardship**

Professionalism

LEGEND

C Constipation

CBT Cognitive Behavioral Therapy

C. Difficile *Clostridium difficile* infection

D Diarrhea

FODMAP Fermentable Oligosaccharides, Disaccharides, Monosaccharides, and Polyols

GI Gastrointestinal

GIT Gastrointestinal Tract

IBD Inflammatory Bowel Disease

IBS Irritable Bowel Syndrome

M Mixed

MOA Mechanism of Action

PEG Polyethylene Glycol

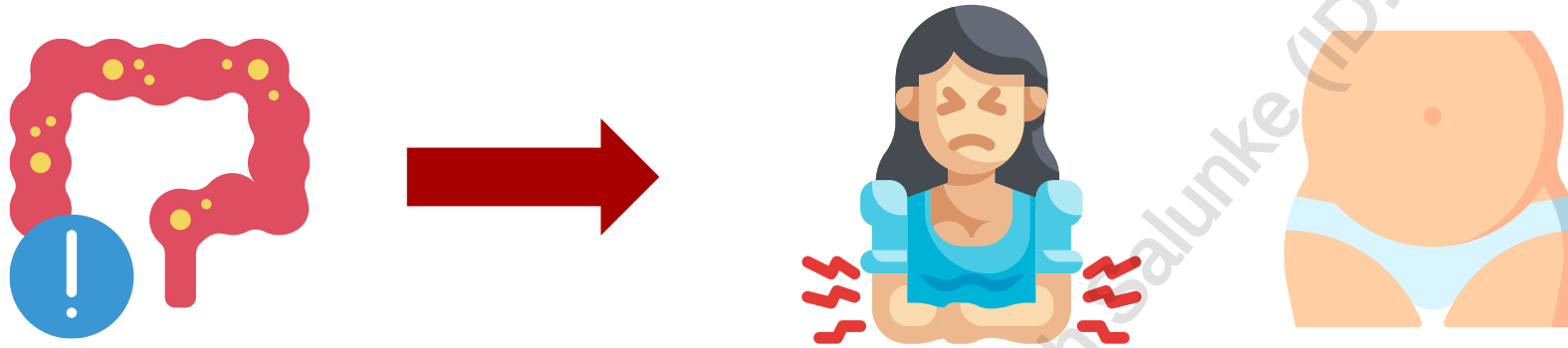
PO Per Os (Oral)

SSRIs Selective Serotonin Reuptake Inhibitors

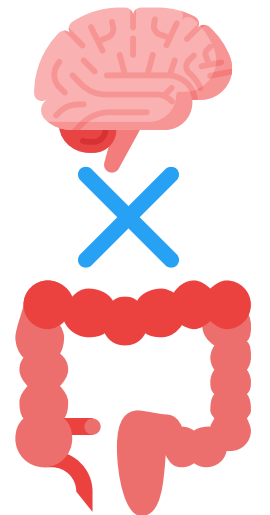
TCAs Tricyclic Antidepressants

U Unclassified

INTRODUCTION AND PATHOPHYSIOLOGY



- Recurrent abdominal pain and altered bowel habits (constipation, diarrhea, or both) that is often associated with bloating
- The cause is not directly understood and is thought to be multifactorial and due to a “disorder” of gut-brain interaction
- Possible causes include: inflammation, stress, genetic, changes in environment, dietary sensitivities, altered intestinal microflora



DIAGNOSIS

FYI

ROME IV Criteria*

- Recurrent abdominal pain for ≥ 1 day per week for ≥ 3 months and associated with ≥ 2 of the following:
 - Related to defecation (the passing of stool)
 - Associated with a change in stool frequency
 - Associated with a change in stool appearance

*Criteria fulfilled for the last 3 months with symptom onset at least 6 months prior to diagnosis

CLINICAL PRESENTATION

Classifications

1. **IBS-C:** constipation predominates
2. **IBS-D:** diarrhea predominates
3. **IBS-M:** mixed
4. **IBS-U:** unclassified

Patients frequently cycle through these phases. The key symptoms of abdominal pain and discomfort are present in all classifications

Symptoms

- Constipation and/or diarrhea
- Abdominal pain, bloating and gas
- Altered appearance of stool results from altered GI motility (can be determined using **Bristol Stool Form Scale**)

Type	Stool Form (Bristol Stool Form Scale)
1	Separate hard lumps like nuts, hard to pass, slow transit
2	Sausage-shaped but lumpy
3	Sausage-shaped but cracks on its surface
4	Sausage-shaped, smooth and soft
5	Soft blobs with clear cut edges
6	Fluffy pieces with ragged edges, a mushy stool
7	Watery, no solid pieces, result of fast transit

RED FLAGS

- Onset age > 50 years old
- Unexplained weight loss
- Rectal masses or bleeding
- Blood in stool (melena)
- Nocturnal symptoms
- Recent history of antibiotic use
- Elevated inflammatory markers
- Family history of colorectal carcinoma, IBD or celiac disease



REFER!



NON-PHARMACOLOGICAL OPTIONS - IMPORTANT

Keep a diary of possible stressors, diet, and symptoms

Treat comorbidities (e.g. depression) and minimize stress

Eliminate triggers (e.g. caffeine, citrus fruit, dairy, cereals, potatoes, fatty foods, alcohol, sugars, fibre)

Low FODMAP diet

Try to exercise regularly

CBT and/or hypnotherapy

Psyllium supplement (soluble fibre)* and/or probiotics

* Insoluble fibre (e.g. wheat bran) is *not* recommended

PHARMACOLOGICAL OPTIONS: IBS-DIARRHEA

Eluxadoline

- MOA: Synthetic opioid receptor modulator
- Improvement in stool consistency
- Should be initiated by gastroenterologist, duration of use limited to 3 months and requires ongoing monitoring
- Adverse effects: constipation, nausea, vomiting
- Contraindicated in patients with biliary duct obstruction, cholecystectomy, alcoholism, pancreatitis, hepatic impairment and chronic or severe constipation

Rifaximin

- MOA: Non-systematic antibiotic targeting gut microbiota
- Improves overall symptoms by altering gut microbiota
- Not Health Canada approved for < 18 years
- Adverse effects include: *C. difficile*-associated diarrhea, nausea
- Duration of use is limited to 14 days, but the course may be repeated twice
- Contraindicated if hypersensitivity to rifaximin or rifamycin derivatives
- Concerns with cost and potential overuse

PHARMACOLOGICAL OPTIONS: IBS-DIARRHEA

RECOMMENDED AGAINST

Continuous loperamide

- Opioid receptor agonist that reduces peristalsis and loss of fluids and electrolytes
- Improves symptoms of diarrhea but not IBS symptoms or abdominal pain
- Adverse effects: abdominal pain, bloating, nausea, vomiting, constipation
- Can be used prophylactically if travelling or if in a situation where diarrhea will worsen (stress, triggers, etc.)

Cholestyramine

- Binds bile acids, forms insoluble complex excreted in feces
- Better efficacy in patients with bile acid malabsorption (lack of data overall)
- Adverse effects: constipation, bleeding

PHARMACOLOGICAL OPTIONS: IBS-CONSTIPATION

Linacotide

- Strong recommendation for improving overall IBS symptoms
- Synthetic peptide that acts on receptors in GIT to cause fluid secretion, increase colonic transit and affect pain sensors
- Not significantly systemically absorbed, no known drug interactions
- Administer once daily **on an empty stomach**, 30 minutes before meals
 - High-fat breakfast can cause loose stools and increase defecation frequency
- Adverse effects: diarrhea, bloating, flatulence, abdominal discomfort

Plecanatide

- Same MOA as linacotide
- Not indicated in < 18 years
- Administer once daily **with or without food**
 - Avoid taking with a high-fat meal; can cause loose stools and increase defecation frequency
- Adverse effects: diarrhea most common

PHARMACOLOGICAL OPTIONS: IBS-CONSTIPATION

RECOMMENDED AGAINST

Osmotic laxatives

- Use is not recommended in managing overall IBS symptoms
- Some evidence to suggest beneficial effects on constipation only
- No improvement in abdominal pain
- PEG is well-tolerated

Prucalopride

- Serotonin receptor agonist, prokinetic effect
- Less evidence of benefit
- Adverse effects: headache, nausea, diarrhea

PHARMACOLOGICAL OPTIONS: IBS

Tricyclic antidepressants - amitriptyline, desipramine, doxepin, imipramine, trimipramine

- Strong recommendation
- Improves symptoms e.g. bloating and abdominal pain
- Adverse effects: drowsiness, dry mouth, constipation (therefore may be better if used in patients with symptoms of diarrhea)
- Used at doses lower than those used in depression

Selective Serotonin Reuptake Inhibitors (SSRIs) - citalopram, fluoxetine, paroxetine

- Conditional recommendation
- Improves symptoms e.g. bloating and abdominal pain
- Adverse effects: increased risk of bleeding, hyponatremia, QT prolongation, sexual dysfunction. May decrease gut transit time, so use in patients with symptoms of constipation
- Used at doses similar to those used in depression

Antispasmodics - dicyclomine, hyoscine butylbromide, pinaverium

- Conditional recommendation
- Anticholinergic/calcium-blocking channel properties, relax smooth muscle
- Use if failed other treatments or use on an as-needed basis
- Adverse effects: dry mouth, dizziness, blurred vision

PREGNANCY AND BREASTFEEDING

IBS-Diarrhea

- Calcium polycarbophil and psyllium are compatible in both pregnancy and breastfeeding
- Colesevelam can be considered in pregnancy if the benefits > risks
- Avoid diphenoxylate with atropine → teratogenic
- Insufficient safety evidence for loperamide, eluxadoline and rifaximin

IBS-Constipation

- Increased dietary fibre, adequate hydration and physical activity
- First-line therapy in pregnancy: Psyllium
- Senna or bisacodyl may be used short-term
- Dicyclomine has been used in pregnancy, inadequate evidence in breastfeeding
- Avoid: linaclotide, prucalopride, pinaverium, trimebutine

TAKEAWAY

- IBS is a multifactorial condition with unclear pathophysiology. It is a clinical diagnosis of exclusion
- Treatment is focused on symptom management (constipation, diarrhea, abdominal pain, bloating)
- Red flags prompt referral
- For IBS-M, treat based on predominant symptom

IBS-D	IBS-C
Pharmacological measures: • Eluxadoline, rifaximin • Loperamide maybe used prophylactically • If pain or bloating → antispasmodics, TCAs	Pharmacological measures: • Linaclotide, plecanatide • Osmotic laxatives can be used to manage constipation symptom but do not improve overall IBS • If pain or bloating → antispasmodics, SSRIs
Nonpharmacological measures should be emphasized in all patients: • Low FODMAP, psyllium, probiotics, psychological therapies, exercise	

CASE 1

HP is a 22-year-old female who presents to her local pharmacy complaining of ongoing gastrointestinal symptoms. She reported that these symptoms started last year when she began graduate school. Most days of the week her stools are very soft and watery. She is also experiencing abdominal pain 2-3 days per week. Her doctor told her this is likely irritable bowel syndrome (IBS) and she is seeking advice from her pharmacist to resolve these symptoms. HP has been diagnosed with anxiety and she takes Alesse® (levonorgestrel 100 mcg/ ethinyl estradiol 20 mcg) PO daily, Synthroid® (levothyroxine) 88 mcg PO daily, and vitamin D 1000 units PO daily. She is allergic to latex (experiences a severe rash). She was previously an avid runner, however, she has not been running much since starting school.

1. What subtype of IBS does HP likely have?
 - a) IBS-D
 - b) IBS-C
 - c) IBS-M
 - d) She does not have IBS



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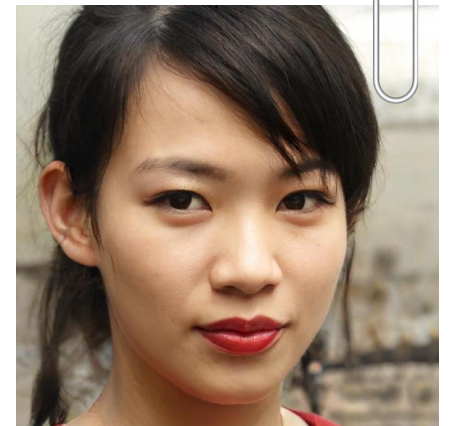
2. Which of the following is NOT an appropriate non-pharm recommendation for HP
- a) Increase physical activity
 - b) Treat comorbid anxiety
 - c) Increase bran intake
 - d) Increase psyllium intake



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3. HP has a long flight next week and she is nervous about her symptoms acting up during the flight. Which of the following is a first-line option for HP on her flight?
- a) Lactulose
 - b) Loperamide
 - c) Rifaximin
 - d) Senna



CASE 1

HP is a 22-year-old female who presents to her local pharmacy complaining of ongoing gastrointestinal symptoms. She reported that these symptoms started last year when she began graduate school. Most days of the week her stools are very soft and watery. She is also experiencing abdominal pain 2-3 days per week. Her doctor told her this is likely irritable bowel syndrome (IBS) and she is seeking advice from her pharmacist to resolve these symptoms. HP has been diagnosed with anxiety and she takes Alesse® (levonorgestrel 100 mcg/ ethinyl estradiol 20 mcg) PO daily, Synthroid® (levothyroxine) 88 mcg PO daily, and vitamin D 1000 units PO daily. She is allergic to latex (experiences a severe rash). She was previously an avid runner, however, she has not been running much since starting school.

4. What would be the best treatment option for HP's anxiety?
- a) Amitriptyline
 - b) Lorazepam
 - c) Venlafaxine
 - d) Propranolol



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CHANGE LOG

August 2025

- Formatting and grammar changes throughout
- Slide 3: Updated legend
- Slide 5: Added footnote, added FYI
- Slides 9, 15: Removed lubiprostone
- Slide 13: Added 'butylbromide' to hyoscine

November 2025

- Formatting and phrasing changes made, slides re-arranged for better flow
- Slide 7: Consolidated and simplified red flags
- Slide 9: Added rifaximin as a treatment option for IBS-D
- Slide 11: Added plecanatide as a treatment option for IBS-C
- Slide 12: Added information to osmotic laxatives
- Slide 14: Updated and consolidated information
- Slide 15: Updated takeaway slide
- Slide 20: Updated references

March 2026

- Slide 2: Updated NAPRA competencies
- Slide 8: Added note on insoluble fibre
- Slide 13: Added strength of recommendation to the 3 listed therapies
- Minor formatting edits made throughout